

Executive Insight from Swimlane Analysis

Scope: Cross-functional workflow across Patient, System, Admin, Doctor, Pharmacy, and Billing

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1. Current Challenges — From As-Is (As-If) Workflow

A. Operational Bottlenecks

1. Admin Verification Bottleneck

- Booking and registration require manual admin validation.
- Patients cannot proceed until admin completes checks → queue buildup.

2. Manual Prescription Handling

- Doctor decision does not flow automatically to pharmacy.
- Manual handoff delays medication preparation.

3. Pharmacy Validation Delay

- Pharmacy re-checks patient and prescription manually.
- Redundant verification slows dispensing.

4. Billing as Final Manual Gate

- Billing and payment are processed manually at the end.
- Entire patient journey cannot close until finance step completes.

Core Pattern: Linear, human-dependent flow → high waiting time.

B. Redundancies

- Multiple validation points: System → Admin → Pharmacy.
- Repeated patient data handling across stages.
- Manual billing despite service data already existing.
- Manual correction loops between patient and admin.

Effect: Rework, longer cycle time, unnecessary process touchpoints.

C. Unclear / Overlapping Responsibilities

- Data correction shared between **Patient and Admin**.
- Validation responsibility split across **System, Admin, Pharmacy**.
- Billing ownership partly manual, partly system-generated.

Organizational Impact: Ambiguity → inconsistent execution → higher exception frequency.

2. Proposed Improvements

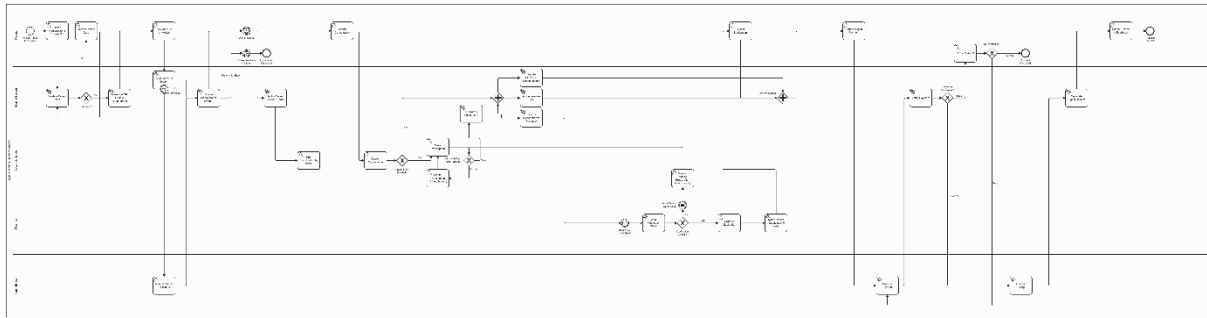


Figure 1. BPMN Swimlane Hospital Process Diagram

The swimlane redesign introduces **system-orchestrated, role-clear, and event-driven workflow**.

A. Clear Role Segmentation

- **System:** Validation, orchestration, EHR recording, auto billing.
- **Admin:** Exception handling only (removed from routine flow).
- **Doctor:** Purely clinical consultation and decision.
- **Pharmacy:** Medication fulfillment only.
- **Patient:** Data submission, visit, payment, feedback.

Improvement: Each lane performs only value-adding work → removes role overlap.

B. Elimination of Redundant Steps

- Single validation at entry point (no repeated checking).
- E-Prescription automatically sent to pharmacy.
- Billing auto-generated from consultation record.
- Pharmacy receives verified data → no revalidation required.

Design Shift: Validate once → reuse across entire workflow.

C. Automated Cross-Lane Orchestration

- System triggers transitions (Registration → Consultation → Pharmacy → Billing).
- QR/auto check-in activates visit automatically.
- Exception handled only when necessary (not default path).

Effect: Removes manual handoffs and waiting gaps.

3. Impact on Efficiency

A. Delay Reduction

- Removes admin verification bottleneck.
- Eliminates prescription handoff lag.
- Billing processed instantly after service.
- Reduces waiting between functional lanes.

Expected Result: Major reduction in total patient cycle time.

B. Improved Communication & Coordination

- System becomes central process coordinator.
- Clear lane ownership removes cross-functional confusion.
- Automated status updates reduce manual follow-ups.

Result: Lower coordination overhead and fewer process errors.

C. Workload Optimization

- Admin workload shifts from routine → exception management.
- Pharmacy focuses on dispensing instead of verification.
- Doctor time fully dedicated to clinical value.

Operational Outcome: Higher productivity without increasing staff.

D. Throughput & Scalability

- Faster patient processing → more patients served per day.
- Reduced dependency on manual processing capacity.
- Standardized digital flow improves predictability and SLA control.

4. Structural Comparison

Dimension	As-Is (As-If)	To-Be (Swimlane BPMN)
Flow Type	Manual & Sequential	Automated & Event-Driven
Validation	Repeated & Fragmented	Single-point
Admin Role	Central bottleneck	Exception only
Prescription	Manual	E-Prescription
Billing	Manual	Automated
Coordination	Human-driven	System-orchestrated
Cycle Time	Long	Significantly shorter
Process Clarity	Ambiguous	Clearly segmented

5. Strategic Insight

The transformation is not merely automation — it is **control architecture redesign**:

- From **human coordination** → **system orchestration**
- From **task repetition** → **data reuse**
- From **role overlap** → **clear ownership**
- From **queue-driven** → **flow-driven**

Business Result

- Faster service delivery
- Reduced operational friction
- Higher reliability and traceability
- Improved patient experience and scalability